KRI 7: Threshold Headroom

Accumulation of productivity significantly far from the threshold, consistently. Draft proposal.

Objective

Identify alert definitions (ADs) where all productive alerts sit well above the monitoring threshold. In such cases the threshold can potentially be raised, so that the non-productive alerts between the current threshold and the lowest productive value are never generated. The result is reduced analyst workload with a known, controlled effect on detection.

The key observation: a threshold raise fails at exactly one point, the lowest productive value. The KRI therefore measures that point directly, rather than a statistical proxy for it.

Step 1: Normalization

All alert values are divided by the threshold:

```
r = alert value / threshold T
```

This places every AD, entity, and parameter type (amount, count, ratio) on one comparable axis where r = 1 is the threshold and r = 2 means twice the threshold. It removes the need for parameter-specific multipliers.

Step 2: The KRI value, Headroom

```
H = q_alpha( productive r-values )
```

H is the alpha-quantile of the normalized productive values.

In plain terms: H answers "by what factor can the threshold be raised while losing at most a fraction alpha of the productive alerts we observed?"

- At alpha = 0, H is simply the lowest productive value. Zero tolerated loss.
- At alpha = 0.05, H is the 5th percentile. Up to 5% of observed productives are accepted as tolerable loss.

alpha is a risk appetite parameter set by the business per scenario risk tier, not a statistical constant. High-risk typologies run at alpha = 0.

Example. An AD has threshold T = 10,000 and its productive alerts over the window have values 24,500 / 31,000 / 38,200 / 55,000. Normalized: 2.45 / 3.10 / 3.82 / 5.50. At alpha = 0, H = 2.45. Every productive sits at least 2.45x above the threshold.

Step 3: Two supporting quantities

```
R = count of non-productive alerts with r < H     (reclaimed volume)
P = count of productive alerts with r < H         (productives at risk)
```

R is the efficiency prize: the false positives that would never have been generated under the raised threshold. Multiplied by average handling time, it converts directly to analyst hours saved.

P is the detection cost: how many observed productives would have been missed. At alpha = 0, P = 0 by construction. This is the number risk and compliance stakeholders need to see explicitly.

Together the three numbers form one decision sentence:

> "Raise the threshold to H x T, eliminate R false positives (Y hours), at P observed productives lost."

Firing condition

```
KRI 7 fires when:  H >= h_min  AND  R >= r_min
```

evaluated on a pooled rolling window (default: last 4 quarters as one sample).

- **h_min** (suggested 1.25 to 1.5): minimum headroom worth acting on. Below this, a raise offers little gain and no safety margin.
- **r_min**: minimum reclaimed volume, set so that the saved hours exceed the cost of running a tuning cycle.

Both parameters have direct operational meaning and can be defended in a governance forum: "we act when we can raise by at least X% and save at least Y hours."

Temporal robustness: instead of requiring conditions to hold in consecutive quarters, H is computed on each half of the pooled window and both halves must clear h_min. Same stability assurance, more data per estimate, faster action.

RAG banding

Band	Condition	Meaning
Green	H < 1.25	Threshold well placed, no action
Amber	1.25 <= H < 1.5	Monitor, re-evaluate next quarter
Red	H >= 1.5 and R >= r_min	Tuning candidate, triggers review

Note: "red" here signals opportunity (avoidable workload), not detection risk.

Small samples

Productives in window (n)	Treatment
n >= 5	Full metric. Report H with a confidence interval; fire on the lower bound.
n = 2 to 4	Indicative only, routed to analyst review.
n = 0 or 1	Routed to a separate retirement / BTL review list. High volume with near-zero yield is the largest efficiency case and is handled through its own action rather than excluded.

Guardrails

1. **Risk tier caps alpha.** High-risk scenarios evaluated at alpha = 0 only.
2. **Recency veto.** A productive near the threshold in the most recent quarter blocks the candidate even if the pooled quantile clears h_min.
3. **Volume floor.** R must clear r_min.
4. **Mandatory BTL validation.** The KRI reads observed above-the-line history and cannot see risk that never alerted. Every fired candidate is validated by below-the-line sampling at the proposed new threshold before any change is implemented.

Positioning: KRI 7 identifies candidates and quantifies the prize. Authorization of any threshold change remains with the existing tuning / BTL process. The KRI is a prioritization instrument, not a threshold-setting model.

Quarterly output

One ranked table per quarter, sorted by reclaimed volume subject to P within appetite:

AD x parameter	H (CI)	New T	R	P	Hours saved	Risk tier	Recommendation
AD-4711 x amount	2.45 (2.1 to 2.8)	24,500	312	0	156	medium	tune-candidate

Recommendations: tune-candidate, retire-review, monitor, insufficient-data.

Summary

Three numbers per AD, computed from data already in the BackTesting pipeline (quantiles and counts, no distributional assumptions):

- **H**: how far the threshold can safely be raised
- **R**: how many false positives disappear
- **P**: how many detections it costs

Each is individually explainable, comparable across all entities and parameter types, and together they turn the quarterly KRI run into a prioritized, risk-gated tuning backlog.